

3D picking for Universal Robots. A URCap whose 3D Pick program node uses a depth camera on the wrist to find every part of the size you enter, number them in the pick order you choose, and move the tool to the grip on each one. Your program opens and closes the gripper.

Planning figures, not guarantees. The numbers below are deliberately conservative estimates derived from the software's limits, its programmed speeds and the camera's noise. This release is a beta and the figures have not been measured on a production cell. **Test with your parts, gripper and layout before you commit to them.**

PERFORMANCE (conservative; verify on your cell)

Smallest part	25 × 25 × 15 mm^E	Length × width × height as it lies. The software accepts 5 mm; depth noise makes that unreliable.
Largest part	200 mm long side^E	The whole part must be in one picture. The software accepts 500 mm.
Size measurement	±5 mm	The noise floor the size match allows for. Parts that differ by less than 10 mm cannot be told apart.
Grip position	±10 mm^E	Plan jaw opening and fingers for this much error. Depends on the camera calibration (4.4 mm RMS in one trial).
Gap between parts	≥ 25 mm^E	The grip check wants 20 mm clear on each side the fingers come down on. Touching parts are skipped.
Cycle time, node only	15 s per part^E	From the start of the node to the tool at the grip, with the closer look on. Excludes your gripper, lift and place.
Throughput	2 parts / min^E	With a short place move. Speeds are fixed and moderate: not a high-rate system.
Motion speeds	0.15 m/s · 36°/s	Linear approach and joint moves, set by the node. Final descent to the grip at 0.05 m/s.
Camera to parts	0.30 to 0.60 m^E	At the picture point. Never closer than 0.25 m.
Area per picture	0.5 × 0.3 m^E	Of table, from 0.5 m. Add picture points (up to 12 per node) for more.
Surface	Flat, level	A taught pick area follows the live table by at most 15 mm.
Size tolerance	5 to 100 %	Default 25 %, never tighter than ±5 mm.
Camera stream	848 × 480, 30 fps	RealSense D435 on USB 3.

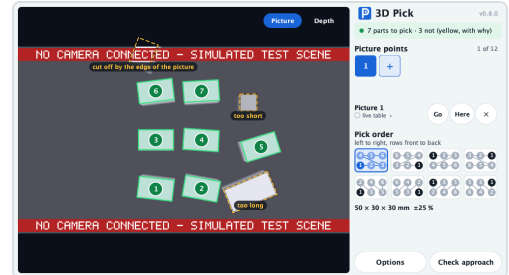
^E Estimate, not a measured or guaranteed value.

NOT FOR

- Bin picking: piled or leaning parts
- Shapes other than boxes and upright cylinders
- Transparent or mirror-like parts
- Telling apart parts of the same size
- Driving the gripper or the place move

CAPABILITIES

- Finds parts by volume, from depth alone:** boxes, and cylinders standing on end. Colour and lighting contrast are not used.
- Any position, any rotation** on the surface. Fingers close across the short side, or the long side of a box if you choose.
- Eight pick orders:** by rows or by columns, from any corner.
- Checks before it moves:** reach by the arm's kinematics, and room for the fingers beside the part.
- Says why not:** rejected parts are drawn in yellow with the reason (off size, touching, out of reach, no finger room, outside the area, cut off).
- Queues what it saw:** the next part is picked without a trip back to the picture point.
- One program node, no child nodes.** Sets `rs_pick_found` and `rs_pick_loc`. Runs in your program; no internet needed.



3D Pick on a PolyScope 5 pendant (simulated scene).

COMPATIBILITY

PolyScope 5	5.4 to 5.26 (e-Series)
PolyScope X	10.8 to 10.14
Robots	UR3e, UR5e, UR7e, UR10e, UR12e, UR16e, UR20
Gripper	Any your program can open and close. The node does not drive it.
WHAT A CELL NEEDS	
Camera	RealSense D435 on a printed bracket between the tool flange and the gripper
Computer	Small Linux computer beside the robot (Raspberry Pi class), on the robot's network
Network	TCP 7621 and 7622, robot to camera computer
Setup	One camera-to-flange calibration; optional pick areas taught with three fingertip touches
URCap	Free, open source (MIT). Kit from Advin.

Download and details: perceptronics.advin.io · **Kit and questions:** nick@advin.io · **Source:** github.com/JimothyJohn/perceptronics

Beta software, provided as is under the MIT license. The integrator is responsible for the risk assessment and safety of the cell. Perceptronics is an independent product, not affiliated with or endorsed by Universal Robots A/S. Universal Robots, PolyScope and URCap are trademarks of Universal Robots A/S; RealSense and other names belong to their owners. © 2026 Nick Armenta, Advin.